

SkillSphere - Setup & Run Guide

COMPLETE STEP-BY-STEP PROCESS TO INSTALL & RUN THE MERN STACK APPLICATION

1. System Requirements & Prerequisites

Before running SkillSphere, ensure the following software tools are installed on your machine:

Component	Required Version	Description / Download
Node.js	v18.x or higher	JavaScript Runtime (nodejs.org)
MongoDB	v6.0 or higher / Atlas	Database Service (mongodb.com)
NPM	v9.x or higher	Node Package Manager (Bundled with Node)

2. Repository Architecture

The application is structured into two clean directories:

- **server/**: Express.js Backend REST API, Mongoose Schemas, Socket.io Gateway
- **client/**: React.js Frontend Single Page Application (Vite)

3. Environment Configuration (.env)

Create or inspect the file server/.env with the following variables:

```
PORT=5000
MONGO_URI=mongodb://localhost:27017/skillsphere
JWT_SECRET=skillsphere_super_secret_jwt_key_2026
CLIENT_URL=http://localhost:5173
NODE_ENV=development
```

4. Step 1: Launch the Express Backend Server

Open a terminal or command prompt in the project root directory and execute:

```
cd server
npm install
npm run dev
```

Expected Output:

MongoDB Connected: localhost

■ SkillSphere Server listening on http://localhost:5000

5. Step 2: Launch the React Frontend Client

Open a second terminal window or command prompt and execute:

```
cd client
npm install
npm run dev
```

Expected Output:

VITE v8.2.1 ready in 500 ms

→ Local: <http://localhost:5173/>

6. Accessing & Testing the Application

- Open <http://localhost:5173/> in your Web Browser.
- Click **Get Started** to register a new account.
- Complete the 3-step onboarding profile.
- Browse match recommendations on the **Matches** page.
- Send swap proposals and test live messaging via **Chat**.

7. Troubleshooting Common Issues

Issue: Error: listen EADDRINUSE :::5000

- Cause: Another process is using port 5000.
- Fix: The server automatically falls back to port 5001, or you can kill the process using port 5000.

Issue: MongoDB Connection Error

- Cause: Local MongoDB service is not started.
- Fix: Start MongoDB service (net start MongoDB on Windows) or update MONGO_URI in server/.env to a free MongoDB Atlas connection string.